

Heading

$$1 = 1$$

$$3 = 1 + 2$$

$$6 = 1 + 2 + 3$$

$$10 = 1 + 2 + 3 + 4$$

$$15 = 1 + 2 + 3 + 4 + 5$$

and by induction on n

$$\frac{n(n+1)}{2} = 1 + 2 + \cdots + n$$